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SUMOBOT

Mechatronic Project 478
Project Proposal

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saam vorentoe

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Name: Chanelle Blanchè Student no: 25929291

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Executive Summary

Title of Project
Sumo Robot (Competition)
Objectives
Design an autonomous Sumobot capable of detecting, pushing and removing an opponent from a sumo arena in a competitive environment (while remaining within boundaries)
What is current practice and what are its limitations?
Current Sumobot designs use simple sensor-based control systems with line and opponent detection. Limitations include limited adaptability to different opponents, inefficient pushing strategies and constraints imposed by size, weight and power regulations.
What is new in this project?
The project will work on improving control strategies, sensor integration and mechanical design to improve the robot's ability to detect opponents, avoid leaving the ring and to apply enough pushing force.
If the project is successful, how will it make a difference?
The project will showcase a Sumobot design that balances sensing, control and mechanical design/performance. The results can provide a reference design and strategy for future robotic competitions and robotics projects. Some parts can be adapted onto various other robotic projects (like the control systems and certain component use)
What are the risks to the project being a success? Why is it expected to be successful?
Potential risks include unreliable sensor readings, insufficient traction or pushing force and software control issues affecting robot behaviour. These risks are handled through iterative testing, careful component selection and systematic design and programming.
What contributions have/will other students made/make?
None will have a direct impact – no prior university robotics projects will be used as reference.
Which aspects of the project will carry on after completion and why?
The robot platform and control software can be further developed to improve strategies, sensing capabilities and mechanical design for future competitions or robotics research.
What arrangements have been/will be made to expedite continuation?
If this project is repeated in the future, careful documentation of my process and development could help other students make different design and strategy decisions.

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1 Introduction

Robotics competitions form an environment where participants can compete against eachother – showcasing their designs and competitive strategies. Sumo-robot competitions are a well-known example. Robots are required to navigate the arena and ‘fight’ against the opposing robot by pushing them out of the boundaries of the ring.

This project focuses specifically on the Autonomous Mini Sumobot class. A robot must be designed and developed to comply with the Robogames competition rules. The robot is expected to use sensors and control/strategy algorithms to detect said opponent, navigate the arena and then apply enough force to counter the opposing robot and push it out of the arena bounds. The most important components of the robot design being speed, momentum, pushing force, traction and precise autonomous strategies.

The project combines knowledge from multiple different engineering tools. The main module knowledge used for this project stems from electronics, computer systems and machine design as well as key skills developed during the E-design module. By combining mechanical design, electronic systems and control strategies, the project will aim to create a functional and competitive robot. The completed robot will then be used in a knock-out style match between students to evaluate its performance.

2 Objectives

As mentioned above, this project aims to design and develop an autonomous sumo robot capable of competing in the Mini Sumo class according to the Robogames competition rules. The objectives of the project are therefore:

Table 1: Objectives

Objective	**Outcome Goal**
2.1. Design a compact sumo robot that complies with the Mini Sumo competition regulations	A functioning robot that complies with Robogames' regulations (10cmx10cm, 500g). The design should include some defensive choices like being sturdy enough to withstand being pushed around
2.2. Develop sensing system that enables the robot to detect opponents and remain within the arena boundaries	Successful boundary detection, successful starting line detection and arena navigation and successful detection of an opponent
2.3 Develop a control strategy that enables the robot to attack its opponent	While still following the guidelines from 2.2, the robot can successfully utilize an offensive strategy to push its opponent from the arena
2.4 Demonstrate the functionality of the robot through testing and participation in the sumo knock-out competition	The Sumobot successfully detects and pushes the other robot from the arena without being pushed out first and while staying within the arena boundaries

3 Motivation

Autonomous robotics combines theoretical knowledge, design and testing in a loop of trial, error and improvement. Such a seemingly simply project requires knowledge of a variety of engineering principles. It integrates mechanical design, the design of electronic systems and the implementation of control and strategy algorithms.

All the above makes the Sumobot project particularly valuable for developing practical engineering and problem-solving skills.

The design of a Mini Sumo robot requires careful consideration of traction, stability, sensing capabilities and control strategies to detect an opponent and successfully shove it out of the arena while remaining within the ring boundaries.

This project makes use of knowledge gained in the Computer systems, Electronics, Control systems, E-design and Machine Design modules. By integrating these areas, the project provides an opportunity to demonstrate the practical application of engineering principles in the design and development of a autonomous robotic system.

4 Planned Activities

The following activities are anticipated for the execution of the project. The costs and time scales associated with each activity are given in Appendix A.

4.1. Review Literature

Study literature regarding the Sumobot competition guidelines as well as literature containing existing strategies and design concepts. Further, review literature that handle certain components and sub-systems in more depth.

4.2. Identify Compatible Design Approaches

Investigate possible design concepts and approaches for a sumo robot that can compete within the competition guidelines. Determine the categories of strategies such as offensive pushing, defensive stability and autonomous detecting behaviour. Evaluate which approaches are the most effective considering the robot size, weight, sensor systems and motor capabilities.

4.3. Compile Design Requirements

Determine the requirements that the sumo robot must meet to comply with competition guidelines and perform effectively in matches. Compile evaluation criteria to compare different design concepts. These criteria will consider mechanical performance, electrical design and performance, control system functionality, ease of manufacturing and maintenance as well as longevity.

4.4. Investigate Concepts for the Robot System

Develop several concept design ideas for the sumo robot including mechanical structure, drive systems, sensor placement, PCB and electronics layout and control strategies. Evaluate each concept using preliminary calculations and design analysis to determine how well they satisfy the requirements defined in Activity 4.3. Compile the most promising concepts based on this assessment.

4.5. Simulate the Most Promising Concepts

Investigate the compiled concepts in more detail. Develop simulations or analytical models to evaluate performance aspects such as robot acceleration, pushing force capability, stability and sensor detection range. Use these simulations to analyze how well each concept satisfies the design requirements and competition guidelines.

4.6. Select Preferred Concept

Use the information obtained from the concept evaluation and simulation to select the most suitable design for the sumo robot. The decision will consider

performance, reliability, cost, manufacturability and compliance with competition constraints.

4.7. Design Review

Present the chosen robot design to the project supervisor for review. Feedback from this review will be used to confirm the design direction and identify potential improvements before fabrication begins.

4.8. Design Prototype System

Develop the detailed design of the sumo robot concept including mechanical components, electrical circuitry, sensor integration and embedded control software. Prepare manufacturing drawings, circuit schematics and system integration plans.

4.9. Manufacture Prototype

Manufacture and assemble the mechanical structure of the robot. Then, integrate the electrical components, sensors and microcontroller system. Ensure that the prototype is constructed according to the design specifications.

4.10. Test Prototype

Test the robot to evaluate its performance in areas such as movement control, sensos, obstacle detection and pushing capability. Conduct tests under conditions similar to competition scenarios – by using a home-made testing arena and some mock-up opponents - and record the results.

4.11. Validate System Performance

Compare the experimental test results with the predicted performance from simulations and calculations. Identify the differences and evaluate whether the design meets the specified requirements.

4.12. Finalise Report

Document the entire project including the literature study, design process, concept evaluation, prototype development, testing procedures and results. Provide conclusions regarding the effectiveness of the developed sumo robot and recommendations for future improvements.

5 Project Risk Assessment

This section considers the main risks that could influence the development of the Sumobot project. The risks include technical challenges, time constraints and performance limitations of the system components.

Table 2: Risk Assessment

Risk	Strategy
5.1 Not enough traction or pushing force (design failure)	Maximise mass through mechanical design to avoid putting unnecessary stress on electrical components. Complete preliminary calculations before performing iterative testing to ensure that it meets set out requirements/ is robust enough
5.2 Sensor accuracy/ faults	Sensors can be noisy, generate some faulty data or even detect incorrectly (as in the case where an object is at just the wrong angle so a sensor doesn't pick it up or picks it up as further away). Some have dead zones close to the sensor transmitter and receiver as well. This will be handled by careful calibration, data processing methods and proper placement of the sensor to avoid general issues like the dead zone. All of this will be confirmed through thorough and continuous testing
5.3 Offensive decision-making errors/ inaccuracies	The system might overshoot or not be as mechanically precise as expected from the coded commands. This will be handled by some form of a controller – PID control etc.
5.4 Component failure (electrical)	Some electrical components can be faulty or break during incorrect testing or accidental circuit failure/ shorts. This can be handled by careful planning, testing on a breadboard before implementing it on the circuit and by buying duplicates of some key components.
5.5 Component failure (mechanical)	Mechanical failure mostly stems from faulty design choices in the perspective of this project. This can be mitigated by preparing detailed design calculations and support structures within the design
5.6 Time management issues	Sticking to schedule is the ideal approach, but in the (likely) case of that not happening, it can be managed by optimizing what activities require less time as the

	project progresses and we gather more knowledge regarding what each section might entail

6 Conclusions

This project proposal outlines the design and development of an autonomous Mini Sumo robot capable of competing within the Robogames competition rules. The project combines mechanical design, electronic systems and embedded control algorithms to produce a robotic system capable of detecting an opponent and pushing it out of the arena while remaining within the ring boundaries.

The planned activities describe an engineering design approach that includes literature review, concept development, simulation, prototype manufacturing, testing and design iteration. These steps ensure that the final robot design is based on careful analysis and evaluation rather than trial-and-error alone.

The project provides an opportunity to apply knowledge obtained from several engineering modules, including Computer Systems, Electronics, Control Systems, Machine Design and E-design. By utilizing these disciplines, the project demonstrates the practical implementation and application of engineering theory in a real-world robotics scenario.

Project risks such as sensor reliability, mechanical performance and time constraints have been identified and strategies have been discussed. Through systematic design, testing and refinement, these risks can be managed to ensure successful completion of the project.

If successful, the project will result in a fully functional autonomous Sumobot and will provide valuable insight into the design of compact robotic systems. The outcomes of the project may also serve as a reference for future students and robotics projects involving autonomous sensing, control and mechanical design.

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Appendix A Planning Details

Table A.1: Estimated Cost per Task (Theoretical)

Activity	Engineering Time		Running Costs	Facility Use	Capital Costs	MMW Labour		MMW Material	TOTAL
	hr	R	R	R	R	hr	R	R	R
Review Literature	25	12500							12500
Identify Compatible	30	15000							15000
Compile Design Requirements	20	10000							10000
Investigate Concepts for the Robot System	50	25000		200					25200
Simulate the Most Promising Concepts	60	30000		300					30300
Select Preferred Concept	20	10000							10000
Design Review	5	2500							2500
Design Prototype System	50	25000	800						25800
Manufacture Prototype	60	30000	1800	300		2	700	200	33000
Test Prototype	50	25000	200	300	150				25650
Improve Prototype and Validate System Performance	80	40000	80	200					40280
Finalise Report	100	50000							50000
TOTAL	**550**	**275000**	**2880**	**1300**	**150**	**2**	**700**	**200**	**280230**

Table A.2: Estimated Cost per Task (Real)

Activity	Engineering Time		Running Costs	Facility Use	Capital Costs	MMW Labour		MMW Material	TOTAL
	hr	R	R	R	R	hr	R	R	R
Review Literature	25	0							0
Identify Compatible Design Approaches	30	0							0
Compile Design Requirements	20	0							0
Investigate Concepts for the Robot System	50	0							0
Simulate the Most Promising Concepts	60	0		80					80
Select Preferred Concept	20	0							0
Design Review	5	0							0
Design Prototype System	50	0	700						700
Manufacture Prototype	60	0	1800			2	700	200	2700
Test Prototype	50	0	200		150				350
Improve Prototype and Validate System Performance	80	0	80						80
Finalise Report	100	0							0
TOTAL	**550**	**0**	**2780**	**80**	**150**	**2**	**700**	**200**	**3910**

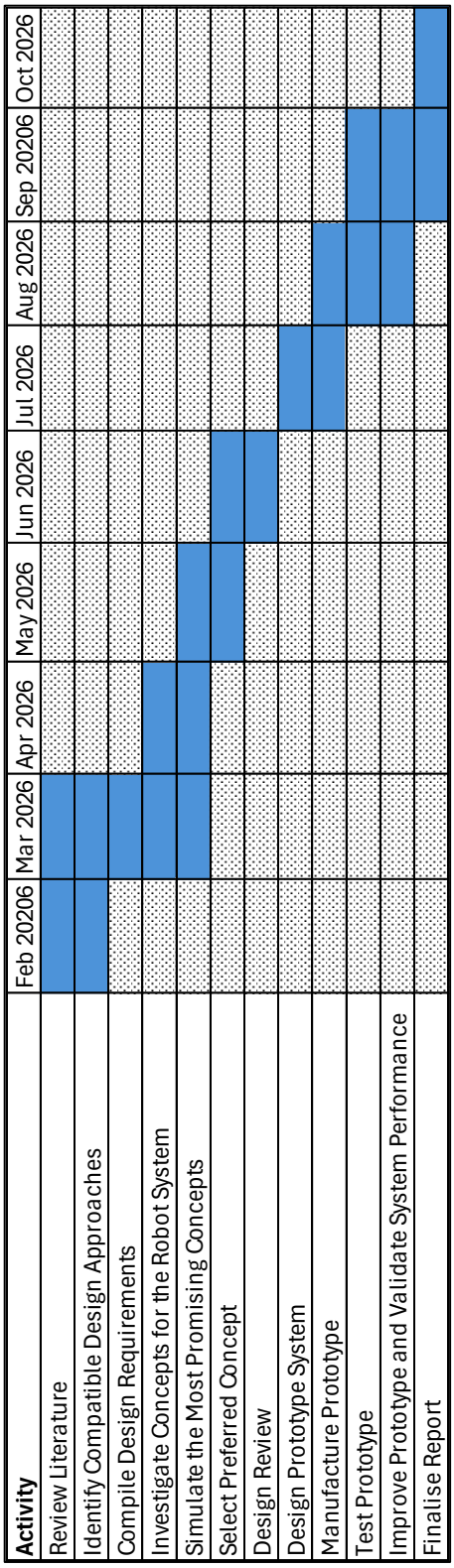


Figure A.1 - Gantt Chart for Sumo Robot Project

Appendix B ECSA Outcome Self-Assessment

Table B.1: ECSA outcome self-assessment

Graduate Attribute	How the project satisfies the attribute	Evidence that it has been satisfied
GA1: Problem solving	The project approaches an open-ended problem: designing an autonomous robot capable of detecting an opponent and pushing it out of the arena while remaining within the ring boundaries. The robot must operate in a competitive environment where strategies, sensing reliability and mechanical performance influence the outcome.	Test results, demonstrations, design calculations, analysis of different solutions and the final knock-out match will serve as evidence that the project problem was analysed and a solution was created
GA2: Application of scientific and engineering knowledge	The project requires the application of engineering principles such as machine design, electronics/ electrical design, control systems and computer systems. This is applied in areas sych as sensor designs, motors, mechanical drive systems and structure design as well as autonomous decision-making algorithms.	Calculations relating to motor torque, traction forces and power requirements, sensor operating principles and justification of design decisions based on engineering theory.
GA3: Engineering design	The project requires the development and synthesis of multiple subsystems into a functional robot. This includes the mechanical structure, motor drive system, sensing hardware and embedded control algorithms. Design iterations will be required to meet performance requirements and competition constraints/ regulations	CAD models of the robot chassis, circuit/PCB schematics, component selection justification, design revisions and documentation of the final integrated robot system
GA5: Use of Engineering Tools	The project will utilize multiple engineering tools throughout its development such as CAD software, microcontroller programming environments, electronic testing equipment and prototyping tools	CAD drawings and assemblies, simulation or modelling results, microcontroller firmware, oscilloscope or sensor testing results, documented use of laboratory tools.
GA6: Professional and technical communication	The project requires technical communication through formal documentation and presentations. The design process, calculations, experimental results and	Project proposal, progress reports, final technical report, diagrams, tables, and verbal explanations of the design decisions during presentations

	conclusions must be clearly communicated in engineering reports	or discussions with the supervisor.
GA8: Individual and multidisciplinary teamwork	The project is done individually but requires knowledge from several engineering disciplines including mechanical design, electronics and embedded control systems. Successful completion requires integration of these fields. The collaborative part could come in when communicating with lab staff and machinists as well as the other sumo-robot skripsie students	Evidence of mechanical design work, electronics integration, embedded software development and system testing
GA9: Independent Learning Ability	The project requires learning beyond what was taught throughout the first few years of engineering. Instead, what was taught should be used as a starting point to develop and learn new skills – such as specific microcontroller setups, new coding languages and IDEs, specific strategies and controllers commonly used for the robot.	Literature review, interpretation of component datasheets, development of control strategies,implementation of new techniques learned independently

Appendix C Responsible Use of Resources and End-of-Life Strategy

The responsible use strategy focuses on selecting components that are readily available as well as avoiding unnecessary manufacturing and designing the robot so that parts can be reused in future projects. The end-of-life strategy considers how components will be reused, recycled or safely disposed of after completion of the project.

Table C.1: Responsible use of resources and end-of-life strategy

Aspect	Responsible-use strategy	End-of-life strategy
Components (Electronic)	Common electronic components like sensors, motor drivers and microcontrollers will be used where possible. Circuits will be tested on breadboards before implementing it on the final circuit board	Functional components can be reused, spare parts can be reused. Broken or damaged components will be disposed of through confirmed e-waste disposal methods.
Components (Mechanical)	Mechanical parts will be designed to minimise material use while still being structurally viable. Modular components will be used to ease assembly and maintenance	Reusable components such as motors, wheels, fasteners and structural elements can be reused in future robotics projects. Damaged mechanical parts will be recycled where possible (or possibly repurposed in some way)
Prototype materials	Prototyping materials such as 3D-printed components will be used mainly for testing and development to reduce manufacturing cost and material waste	Some parts might be reused in the project or modified. Unusable plastic components will be disposed of according to appropriate recycling procedures where available.
Sensors and control hardware	Sensors will be selected and tested to reduce unnecessary replacement or component waste. Proper power regulation and circuit protection will be implemented to avoid component damage	Sensors and electronic boards that remain functional after project completion can be reused in future robotics or laboratory experiments. Or they can be used to test on and improve by future students
Power/ Energy system	The battery system will be selected to safely meet the power requirements of the robot. Proper charging and handling procedures will be followed to prevent damage or safety hazards.	If the battery remains functional it can be reused for other robotics applications. Degraded or damaged batteries will be disposed of through battery recycling facilities.